

# Introduction to Coding (Python)

## Parent / Student Syllabus Handout

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### COURSE SNAPSHOT

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**Length:** 18 weeks (1 hour in-person weekly)

**Primary language:** Python 3

**Audience:** Beginners, no prior experience

**Model:** In-person lesson + weekly take-home practice

### COURSE PURPOSE

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Students build real coding fundamentals - reading, writing, testing, and explaining small Python programs - with an emphasis on clear thinking, persistence, and problem-solving over memorization. By the end of the semester, students can build small command-line programs independently and explain how their code works.

### WHAT STUDENTS WILL LEARN

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- Variables, data types, and expressions
- Conditionals (if/elif/else)
- Loops (for and while)
- Functions and modular code
- Lists, dictionaries, and strings
- Debugging and error reading
- Basic algorithms (search and sort)
- File input/output
- Code comparison across Python, Lua, C++, and Java

### MATERIALS AND PLATFORMS

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- Local Python install with an editor (recommended)
- Browser-based coding environments when needed
- Course resources (Udemy, Learn Python the Hard Way, handouts)
- GitHub for submissions and revision tracking

### WEEKLY STRUCTURE (60 MIN)

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1. Warm-up and review
2. New concept mini-lesson
3. Guided practice
4. Assignment launch and Q&A

*Students complete homework between sessions.*

### HOMEWORK & SUBMISSION

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Weekly homework follows one connected story, **BranchQuest**, building a class text adventure. Each week is submitted on its own **quest branch** via merge request - a gentle first taste of version control.

- One assignment per week
- Submitted through GitHub before the next class
- Commit progressively (not one final upload)
- Students may explain their code briefly in class

### MAJOR COURSE COMPONENTS

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- Weekly coding homework (individual)
- Collaborative class game project - a Python text-based "choose your adventure" built in small teams
- Final individual coding project
- One research paper on a technology topic

### RESEARCH PAPER OVERVIEW

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One paper (**minimum 1,000 words**) with milestone checkpoints across the semester:

- topic proposal
- bibliography
- thesis and outline
- draft
- final paper + short presentation

### ACADEMIC INTEGRITY & AI POLICY

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- **AI-generated code/content is not allowed** for homework or projects.
- Students must be able to explain submitted work.
- Commit history and in-class walkthroughs verify authorship.
- Students sign the Academic Honor Statement in Week 1.

### FAMILY SUPPORT

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- Encourage a consistent weekly homework routine
- Ask students to explain what they built
- Celebrate progress, persistence, and debugging effort
- Ensure on-time submission through GitHub

### SUGGESTED GRADING

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- **Weekly homework** - 40%
- **Collaborative game project** - 25%
- **Final individual project** - 20%
- **Research paper** - 10%
- **Participation / professional habits** - 5%

### COMMUNICATION AND HELP

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Students should bring questions to class each week; debugging confusion is normal, and asking clear questions is part of learning. Parents/guardians are welcome to contact the instructor about pacing, workload, or support needs.